

Deep Learning Approaches for Pneumonia Detection: A Comparative Study on Chest X-Ray Images

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Abstract—Artificial intelligence has revolutionized medical image classification, enabling faster and more accurate diagnoses through deep learning approaches. Early pneumonia detection is crucial for improving patient outcomes, yet remains challenging for clinicians. This paper presents a comprehensive evaluation of deep learning architectures for pneumonia detection using chest X-ray images from the MedMNIST dataset [1]. We systematically compare the performance of four CNN architectures (ResNet-18, ResNet-50, AlexNet, and DenseNet-169) with varying hyperparameters, focusing on the critical balance between false negative rates and false positive rates—a key consideration in medical diagnostics. Our experiments demonstrate that ResNet-50 with data augmentation achieves optimal performance with an AUC of 0.979 and 0% false negative rate, while the simpler AlexNet architecture provides competitive results with lower computational demands. Data augmentation improved model performance by an average of 3.4% for AUC scores, with particularly strong effects on ResNet architectures. Our implementations outperform previous MedMNIST benchmarks for similar architectures, with ResNet models showing significant improvements in both accuracy and AUC metrics. We identify and analyze key trade-offs between performance, computational efficiency, and error types to provide practical recommendations for implementing pneumonia detection systems in clinical settings.

Index Terms—Deep Learning, Medical Image Classification, Pneumonia Detection, False Negative Rate Optimization, ResNet, AlexNet, DenseNet, Data Augmentation, FNR-FPR Trade-off, Transfer Learning, Chest X-ray Analysis

I. INTRODUCTION

In this section we first review the clinical importance of automated chest X-ray analysis, then formalize the specific problem of binary pneumonia detection, and finally outline our objectives and the paper’s organization.

I-A BACKGROUND

We begin by situating our work in the broader context of AI-enabled medical image analysis.

Medical image classification has become essential in modern healthcare, largely due to advancements in AI and machine learning that enable early detection of diseases, thus allowing more efficient treatments and reduce clinicians’ workloads [5].

In this study, we focus on the early detection of pneumonia cases via thorax X-ray images retrieved from the official dataset [2] like the ones seen in Figure 1.

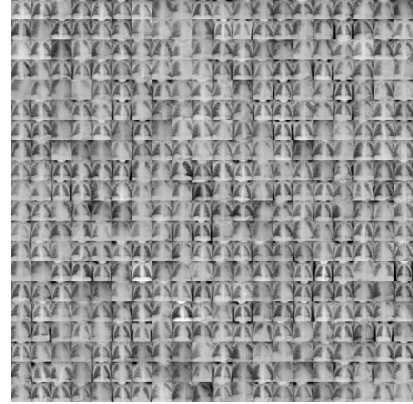


Fig. 1. Overview of Chest X-Ray Images from the chosen dataset

However, pneumonia detection from chest X-rays has many challenges, namely:

- **Variability in Presentation:** Pneumonia can manifest differently depending on its severity, pathogen, and stage. Its signs often overlap with other conditions like congestive heart failure or chronic obstructive pulmonary disease [7].
- **Detection Challenges:** The disease is difficult to identify when complicated by lung conditions like fluid overload (pulmonary edema), bleeding, volume loss (atelectasis or collapse), lung cancer, or post-radiation/surgical changes.
- **Image Quality Issues:** Poor quality or ambiguous images, particularly in early or mild cases, can further complicate diagnosis.
- **Comparative Diagnostic Limitations:** While comparing current chest X-rays with previous ones (when available) and evaluating clinical history can enhance diagnostic accuracy, many pneumonia patients are imaged for the first time [6].

Pneumonia remains a major public health concern, accounting for over 15% of deaths in children under 5 worldwide and ranking among the top 10 causes of death in the US as of 2015 [6]. Figure 2 shows a snapshot from an interactive graph from Our World in Data [8] that provides interactive charts by year, region, and even animated world maps. The tool also allows image export and offers detailed insights into the disease trends and underlying causes.

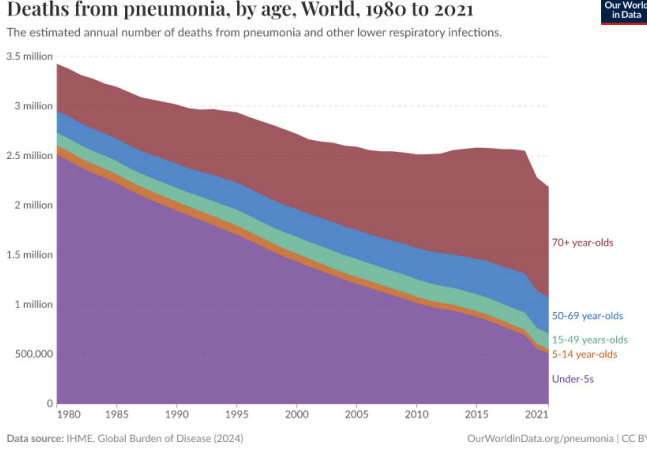


Fig. 2. Evolution of Deaths by Pneumonia

Doctors do not always agree on pneumonia predictions; one study reported a 96% negative predictive value but only a 29% sensitivity for clinical assessments [9]. This variability underscores the need for additional diagnostic support, motivating our proposal of a deep learning approach for pneumonia detection.

I-B PROBLEM STATEMENT AND MOTIVATION

Although we have talked about why pneumonia is important, now we formalize the open issues our study tackles.

Despite the potential of deep learning classification in medical imaging, several challenges persist [4]:

- Limited generalizability of models across different imaging conditions.
- Variability in image quality and acquisition protocols.
- Need for robust performance with limited training data.
- Computational efficiency of classification models.

Our research addresses these challenges by:

- Evaluating multiple pre-trained network architectures.
- Investigating the impact of data augmentation techniques.
- Providing a comprehensive comparative analysis of transfer learning approaches in pneumonia detection.

I-C OBJECTIVES AND CONTRIBUTIONS

To address the challenges outlined above, this work pursues the following objectives and makes these contributions:

- 1) Assess the performance of pre-trained models (such as ResNet, AlexNet, and DenseNet) architectures for pneumonia classification.
- 2) Quantify the impact of data augmentation on model generalization and accuracy (as well as other metrics we found most adequate).
- 3) Provide insights into transfer learning strategies for medical image classification.
- 4) Benchmark our approaches against the existing MedMNIST classification results [1], as well as similar datasets, as shown in Section II.

Our contributions along this paper are:

- A systematic evaluation of deep learning models for binary pneumonia classification.
- An empirical analysis of data augmentation effectiveness.
- Recommendations for model selection and transfer learning in medical imaging.

I-D ORGANIZATION

This paper is organized as follows:

- Section I introduces some background, and states our motivation and objectives.
- Section II reviews related work in medical image classification.
- Section III describes our dataset and preprocessing methodology.
- Section IV details our experimental setup and model architectures.
- Section V presents and discusses our experimental results.
- Section VI concludes the study and suggests future research directions.

II. RELATED WORK

This section reviews relevant literature on deep learning approaches for pneumonia detection using chest X-ray images. We first examine key datasets and standardization efforts that have facilitated research in this domain. Then, we explore model development approaches and methodological refinements proposed by various researchers. Finally, we analyze the gaps in current approaches to position our research contribution.

II-A LITERATURE REVIEW ON MEDICAL IMAGE CLASSIFICATION FOR PNEUMONIA DETECTION

Firstly, we survey the datasets and standardization benchmarks that establish pneumonia detection research. We then review leading model architectures—both classic CNNs and recent transformer-based approaches—and conclude by identifying remaining methodological and clinical gaps.

II-A1 Datasets and Standardization Efforts

Robust model development relies on accessible, well-annotated datasets and agreed-upon data splits. We summarize the most widely adopted chest X-ray collections below.

MedMNIST (Standardized Datasets) [1] provides 18 standardized datasets for 2D and 3D biomedical image classification with multiple size options (28, 64, 128, and 224) [2]. This collection facilitates reproducibility through standard train-validation-test splits and enables consistent evaluation between different models. MedMNIST focuses primarily on the machine learning aspects rather than end-to-end systems, creating a foundation for benchmarking with standardized datasets across various medical classification tasks including pneumonia detection.

Kaggle Chest X-ray Pneumonia Dataset [17] can be considered another resource for pneumonia detection research. The dataset contains anterior-posterior chest X-rays from pediatric patients (aged 1-5) from Guangzhou Women and Children’s Medical Center. All images underwent quality control to remove low-quality scans and were diagnosed by two expert physicians. Beyond binary classification (Normal vs. Pneumonia), the dataset allows differentiation between bacterial and viral pneumonia, making it valuable for more nuanced detection models.

II-A2 Model Development and Approaches

Researchers have proposed a range of network architectures, ranging from deep CNNs to novel transformer models—to tackle pneumonia classification. Below we list two very important milestones.

CheXNet (Rajpurkar et al.) [16] trained on the ChestX-ray14 dataset containing 112,120 frontal-view X-rays with 14 pathologies (Wang et al. 2017) and evaluated against four radiologists (avg. F1 0.387 vs. CheXNet F1 0.435) [16]. It represented a significant advancement by demonstrating that deep learning models could match or exceed radiologists’ performance in pneumonia detection. The study introduced DenseNet-121 and DenseNet-169 architectures trained on the extensive ChestX-ray14 dataset and established transfer learning with ImageNet weights as an effective strategy for medical image classification tasks.

Vision Transformers for Medical Diagnosis (Zhang et al.) [20] pushed the field forward by adapting Vision Transformers (ViTs) specifically for pneumonia detection. Their work demonstrated that with appropriate modifications, transformer architectures could outperform traditional CNNs in clinical diagnostic tasks, suggesting new directions beyond convolutional approaches.

II-A3 Methodology Refinement

Beyond raw model accuracy, several studies have focused on improving robustness through class balancing and advanced preprocessing. We highlight these methodological refinements here.

PLOS ONE Study on Unbalanced Classes [18] addressed a critical challenge in medical image datasets—class imbalance. The authors systematically explored various balancing strategies including oversampling, undersampling, and weighted loss functions. Their quantitative analysis provides

valuable insights into how these approaches affect model performance on imbalanced medical datasets, as they report that applying a weighted focal-loss increased average AUC from 0.882 to 0.912 (+3.0%) on pneumonia classification.

PMC Study on CNN Optimization [19] focused on image preprocessing techniques like lung segmentation, contrast enhancement, and noise removal. Their comprehensive comparison of approximately 25 different methods offers a reference point for preprocessing strategies in pneumonia detection. The study evaluated various metrics including AUC, F1, and MSE across different approaches. Their best preprocessing pipeline—lung segmentation + CLAHE + Gaussian denoising—boosted pneumonia AUC from 0.89 to 0.96 and F1 from 0.85 to 0.91.

II-B PRE-TRAINED MODELS IN MEDICAL IMAGE CLASSIFICATION

Transfer learning from large natural-image datasets has become standard practice. We review the performance of popular pretrained backbones in pneumonia detection.

II-B1 Convolutional Neural Networks (CNNs)

Deep convolutional models remain the backbone of medical image analysis. Below we review the most widely adopted CNN architectures for pneumonia detection.

ResNet Architectures (ResNet-18, ResNet-50) have demonstrated robust performance across multiple studies, particularly in the MedMNIST benchmarks where they achieved AUC between 94-96% and accuracy between 85-88%. The PLOS ONE study incorporated ResNet-18 as part of an ensemble approach, while several implementations using the Kaggle dataset made use of ResNet variants with transfer learning.

DenseNet Architectures featured prominently in CheXNet, where DenseNet-121 and DenseNet-169 were applied with transfer learning from ImageNet weights. The DenseNet-169 variant emerged as particularly effective for feature extraction in pneumonia detection tasks. The PLOS ONE study also included DenseNet-121 in their model ensemble, demonstrating its continued relevance in medical imaging.

VGG and MobileNet appear frequently in implementations using the Kaggle dataset. These architectures provide different trade-offs between complexity and performance, with MobileNet offering a more lightweight option suitable for deployment in resource-constrained environments.

EfficientNet implementations, particularly EfficientNetB0 as documented in a Kaggle notebook [21], showcase parameter efficiency while maintaining high performance. The approach of initially freezing convolutional layers while training only the classifier head, followed by fine-tuning the entire network after initial head training yielded a +2.1% AUC gain in Esterlita (2022), proving effectiveness for transfer learning in pneumonia detection.

II-B2 Advanced Architectures and Ensembles

Beyond single-model CNNs, recent work has explored hybrid and non-convolutional approaches. Here we summarize these advanced architectures and ensemble strategies.

Vision Transformers (ViTs) represent the cutting edge in medical image classification as demonstrated in the Nature study. These transformer-based architectures offer an alternative approach to the convolutional paradigm that has dominated the field, potentially capturing different types of features from X-ray images. In Zhang et al. (2024) [20], a ViT achieved 97.61% accuracy, 95.0% sensitivity, and 98.1% specificity on the same Kaggle pediatric X-ray set, outperforming ResNet-50 by around 2% overall.

Ensemble Methods have shown particular promise in the PLOS ONE study, which combined GoogLeNet, ResNet-18, and DenseNet-121 using weighted averaging. In PLOS ONE, an ensemble of GoogLeNet + ResNet-18 + DenseNet-121 achieved AUC 0.962 vs. best single-model 0.948 (+1.4%). This approach reduced individual model variability and increased generalization capability, which is especially important for clinical applications where reliability is crucial.

II-C GAP ANALYSIS IN CURRENT APPROACHES

II-C1 Evaluation Limitations

While many studies report high overall accuracy, narrow metric choices can mask critical weaknesses. Here we highlight common evaluation blind spots.

Narrow Metric Focus: Most studies emphasize accuracy as the primary performance indicator, neglecting other important metrics such as interpretability, robustness to noise, or data bias. More comprehensive evaluation frameworks incorporating F1-score, precision, recall, confusion matrices, and AUC would provide a more complete assessment of model performance in clinical contexts.

Limited Generalizability Assessment: Studies often fail to evaluate models across diverse datasets or population demographics. The Kaggle dataset, while valuable, is limited in size and population diversity, potentially restraining model generalization to different clinical settings or patient populations.

II-C2 Methodological Gaps

Existing pipelines often focus on raw performance gains, overlooking practical considerations — this subsection pinpoints main methodological shortfalls.

Class Imbalance Handling: Despite the PLOS ONE study’s focus on this issue, many approaches including CheXNet do not adequately address the inherent class imbalance in medical datasets. More sophisticated techniques like focal loss or synthetic data augmentation via generative models remain underexplored.

Preprocessing Optimization: While the PMC study highlighted the importance of preprocessing, there is insufficient attention to the computational costs of these steps or potential overfitting risks with heavily processed data. A balanced

approach that considers both performance gains and practical implementation constraints is needed.

II-C3 Clinical Integration Challenges

Translating technical models into real-world tools faces its own hurdles. We conclude by discussing the main barriers to clinical adoption.

Interpretability Concerns: Many high-performing models function as “black boxes,” limiting their utility and acceptance in clinical settings. Few studies adequately address how their approaches could be made interpretable for healthcare practitioners.

Clinical Validation: Most works lack rigorous clinical evaluation or sensitivity analysis across different patient subgroups, such as age categories or disease severity levels. This gap between technical performance and clinical applicability represents a significant barrier to real-world implementation.

III. DATA DESCRIPTION AND PREPROCESSING

In this section, we detail the PneumoniaMNIST dataset characteristics, visualize key properties, and describe all pre-processing and augmentation steps applied prior to model training.

III-A DATASET OVERVIEW

The project utilizes a chest X-ray image dataset for pneumonia detection. Specifically, we utilize the PneumoniaMNIST subset of MedMNIST, which comprises 5,856 pediatric chest X-ray images already split into train/validation/test sets by the authors [1], [2]. A better overview of both the overall composition of the dataset and class distribution can be seen in Table I.

TABLE I
DATASET COMPOSITION AND CLASS DISTRIBUTION

Set	Total Images	Normal	Pneumonia	Percentage
Training	4708	1214	3494	80.4%
Validation	524	135	389	8.9%
Testing	624	234	390	10.7%
Total	5856	1583	4273	100%

The dataset exhibits a slight class imbalance, with a skew towards one class (Pneumonia corresponds to around 73% of the total examples — a level of imbalance that prior work has shown can degrade classification performance without corrective measures [18]), which can potentially support the need to use some form of class weighting or data augmentation strategies.

The researchers standardized all images to a consistent size of 28×28 pixels using a center-crop-then-cubic-spline interpolation pipeline as described in Yang et al. [1] to create an MNIST-like format. For the PneumoniaMNIST specifically, the source images were grayscale with varying dimensions ranging from 384×127 to 2,916×2,713 pixels. To handle this variability, they first center-cropped the images using a window size equal to the length of the short edge of each image,

creating a square image that preserved the central region of interest. Only after this center-cropping step did they resize the resulting square images to the final $1 \times 28 \times 28$ pixel format using cubic spline interpolation, which maintains smoother transitions between pixels compared to nearest-neighbor or linear interpolation methods.

III-B DATA VISUALIZATION

Even though the images are available in various resolutions, namely: 28×28 , 64×64 , 128×128 , and 224×224 , we follow the authors' recommendation that 28×28 inputs yield similar or slightly better AUC (+1–3%) while reducing computational cost [1].

To provide intuition about image quality and class distinctions, we plot sample X-rays below. Figure 3 shows the first 10 images from the training dataset with the corresponding labels (0 for Normal and 1 for Pneumonia).

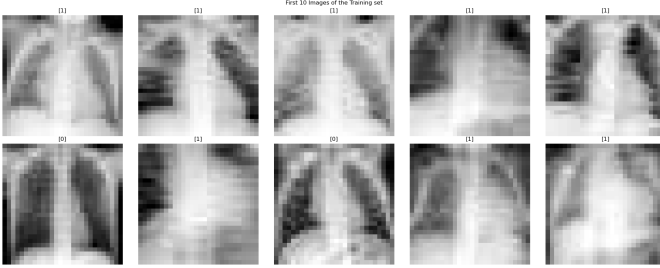


Fig. 3. Sample Chest X-Ray Images from the Training Dataset

III-C PREPROCESSING STEPS

Prior to feeding data into our CNNs, we apply a consistent preprocessing pipeline to all subsets, as summarized in Algorithm 1.

Algorithm 1 Data Preprocessing Pipeline for Chest X-Ray Images

```

1: procedure PREPROCESSDATA(images, labels)
2:   reshaped_images  $\leftarrow$  Reshape(images,  $[n, 28, 28, 1]$ )  $\triangleright$  Add channel dimension
3:   normalized_images  $\leftarrow$  images/255.0  $\triangleright$  Normalize to  $[0, 1]$  range
4:   encoded_labels  $\leftarrow$  OneHotEncode(labels, 2)  $\triangleright$  Convert to categorical
5:   return normalized_images, encoded_labels
6: end procedure

```

The preprocessing steps can be detailed as follows:

- **Image Reshaping:** Images were transformed to a consistent 28×28 pixel dimension with a single channel, preparing them for convolutional neural network input.
- **Image Normalization:** Pixel values were scaled to the range $[0, 1]$ by dividing the original pixel intensities by 255, which helps in faster convergence during model training.

- **Label Encoding:** Original labels were converted to categorical (one-hot) encoding, creating a binary representation suitable for classification tasks.

III-D DATA AUGMENTATION

To mitigate the risk of overfitting and enhance model generalization, we implemented a data augmentation strategy, inspired by common practices in medical imaging to simulate realistic variability [18], [19], as detailed in Algorithm 2.

Algorithm 2 Data Augmentation Pipeline for Chest X-Ray Images

```

1: procedure DATAAUGMENTATION(images, labels)
2:   aug_pipeline  $\leftarrow$  Initialize Sequential Augmentation
3:   aug_pipeline.Add(RandomRotation(0.08))  $\triangleright$  Rotate  $\sim 15$  degrees
4:   aug_pipeline.Add(RandomTranslation(0.1, 0.1))  $\triangleright$  Shift images
5:   aug_pipeline.Add(RandomZoom(0.1))  $\triangleright$  Scale variations
6:   aug_pipeline.Add(RandomContrast(0.1))  $\triangleright$  Contrast adjust
7:   return aug_pipeline
8: end procedure

```

By applying the augmentation steps outlined in Algorithm 2, we generate multiple variations of the original training images. The result can be observed in Figure 4, which compares the original and augmented versions of 5 different images.

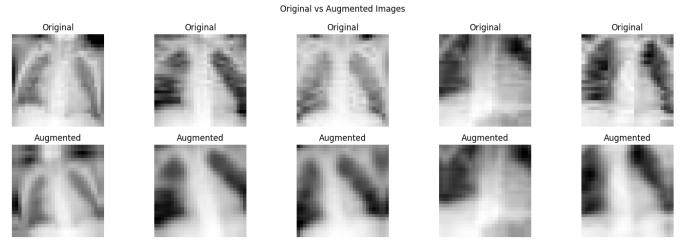


Fig. 4. Original vs Augmented Images

As recommended to prevent information leakage, no augmentation is applied to validation or test sets [3], so as to maintain their integrity.

IV. METHODOLOGY

This section details our experimental approach for pneumonia detection using deep learning models. We first present the overall pipeline architecture, followed by detailed descriptions of the model architectures implemented in this study. We then describe our training strategy, including transfer learning approaches and regularization techniques. Finally, we outline our experimental setup and evaluation metrics used to assess model performance.

IV-A OVERALL PIPELINE

Our project follows a simple two-stage flow: data preprocessing and classification.

Data Preprocessing: Detailed in Section III, this stage involves:

- Reshaping images to a consistent 28×28 resolution.
- Normalizing pixel values to the $[0, 1]$ range.
- Encoding labels to a one-hot categorical format.
- Applying data augmentation (random rotation, translation, zoom, and contrast adjustments) exclusively to the training set to improve model generalization.

Classification: We implemented and trained several convolutional neural network architectures: AlexNet, ResNet-18, ResNet-50, and DenseNet-169. Architectures were selected based on prior success in medical imaging (e.g., Rajpurkar et al. 2017; Yang et al. 2021) [1], [16], and also from classes by our course instructor.

Among these, only DenseNet-169 utilized true transfer learning with pre-trained ImageNet weights, while the other architectures were implemented from scratch following their standard designs but adapted for our smaller input size. Performance metrics were collected and compared between experiments with and without data augmentation, with the best-performing model saved for further use.

IV-B MODEL ARCHITECTURE

We made use of established convolutional neural network architectures including AlexNet, ResNet-18, ResNet-50, and DenseNet-169 for our pneumonia detection task. Each architecture was carefully adapted to work with our dataset of 28×28 grayscale chest X-ray images.

IV-B1 AlexNet

AlexNet is a pioneering CNN architecture that consists of 5 convolutional and 3 fully connected layers. For our implementation, we adapted AlexNet to work with our smaller input images:

- Five convolutional blocks with batch normalization and ReLU activation
- Three max-pooling layers
- Two dense layers with 4096 neurons each, with batch normalization and dropout
- Final classification layer with softmax activation

Due to our smaller input size, we reduced stride sizes and used appropriate padding to maintain spatial information through the network. Additionally, the pooling layers were used to perform the maximum pooling function, and the input size was fixed due to the presence of fully connected layers [11].

IV-B2 Residual Networks

Models in the ResNet family learn residual functions with reference to the layer inputs, rather than learning unreferenced functions. Instead of hoping that stacked layers will directly fit a desired underlying mapping, ResNets let these layers fit a residual mapping [10].

ResNets are known for their residual shortcut connections, which help mitigate the vanishing gradient problem in very deep networks. We implemented both ResNet-18 and ResNet-50 variants:

a) ResNet-18

We adapted the standard architecture for our 28×28 input images by modifying the initial layers. Instead of using a 7×7 kernel with stride 2 as in the original implementation, we used a 3×3 kernel with stride 1 to preserve spatial information in the small input. The model consists of:

- Initial convolution layer (3×3 kernel, stride 1)
- Four stages with two residual blocks each, with increasing filter sizes (64, 128, 256, 512)
- Global average pooling layer
- Dropout layer for regularization
- Dense layer with softmax activation for binary classification (Normal vs. Pneumonia)

b) ResNet-50

This deeper variant uses bottleneck blocks instead of standard residual blocks. Each bottleneck block consists of three convolutions (1×1 to reduce dimensions, 3×3 for feature extraction, and 1×1 to restore dimensions), enabling deeper networks with controlled computational complexity. Our implementation follows the same adaptations made for ResNet-18, with the following structure:

- Initial convolution layer (3×3 kernel, stride 1)
- Four stages with varying numbers of bottleneck blocks (3, 4, 6, and 3)
- Global average pooling, dropout, and final classification layers

IV-B3 DenseNet-169

DenseNet architectures introduce dense connections between layers, where each layer receives feature maps from all preceding layers. This encourages feature reuse and strengthens gradient flow. Our DenseNet-169 implementation addressed the challenge of small input images through:

- A channel expansion layer converting single-channel grayscale to 3-channel input
- Resizing to 224×224 using a resize layer to match DenseNet's expected input
- Leveraging pre-trained ImageNet weights
- Custom classification head with global average pooling, batch normalization, and dropout

For true transfer learning, we initialize with ImageNet-pretrained weights and replace the classifier head, as recommended in Pan and Yang 2010 [12].

IV-C TRAINING STRATEGY

This subsection covers our choices around transfer learning, fine-tuning, and regularization to mitigate overfitting.

IV-C1 Transfer Learning and Fine-Tuning

Transfer learning is a machine learning technique where we reuse knowledge gained from one task or data set to improve model performance on another related task and/or different dataset [12]. For ResNet-18, ResNet-50, and AlexNet, we implemented our models from scratch but with architectures proven on large-scale image classification tasks. For DenseNet-169, we used a true transfer learning approach by directly leveraging pre-trained ImageNet weights.

Although transfer learning typically involves a two-phase approach (first freezing convolutional layers while training only the classification head, then unfreezing selected layers for fine-tuning), our implementation trained all models end-to-end from the beginning, (see Smith et al. 2020 for a comparison of end-to-end vs. phased fine-tuning) [13]. This approach was chosen for implementation simplicity, though a phased fine-tuning strategy could potentially yield better results and represents an area for future improvement.

IV-C2 Regularization Techniques

To prevent overfitting, we implemented several regularization strategies:

- **Batch Normalization:** Each convolutional layer is followed by batch normalization to stabilize training and accelerate convergence.
- **Dropout:** We employed dropout layers with either a fixed rate (0.3-0.5) or a custom scheduler that gradually reduced the dropout rate from 0.5 to 0.3 over the course of training.
- **Early Stopping:** Training was halted when validation performance stopped improving, with a patience of 10 epochs.
- **Learning Rate Reduction:** The learning rate was automatically reduced by a factor of 0.2 when validation loss plateaued for 5 epochs.

IV-C3 Optimization Strategy

All models were trained using standard cross-entropy loss and the Adam optimizer, with hyperparameters chosen to balance convergence speed and stability, below we list our configuration:

- **Loss Function:** Binary cross-entropy
- **Optimizer:** Adam with initial learning rate 1×10^{-3}
- **Batch Size:** {32, 64, 96, 100}
- **Dropout Strategy:** Fixed rates {0.3, 0.5} or scheduled (0.5→0.3); DenseNet-169 used only fixed dropout
- **Data Augmentation:** Applied vs. not applied (training set only)
- **Maximum Epochs:** 100 (with early stopping)
- **Early Stopping Patience:** 10 epochs
- **Model Selection Criterion:** Validation AUC

TABLE II
HYPERPARAMETER SETTINGS AND RANGES EXPLORED

Hyperparameter	Values Explored
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Initial Learning Rate	1×10^{-3}
LR Reduction Factor	0.2 (on validation plateau)
Batch Size	32, 64, 96, 100
Dropout Strategy	Fixed (0.3, 0.5); Scheduled (0.5→0.3)
Data Augmentation	On / Off (training set only)
Maximum Epochs	100
Early Stopping Patience	10
Model Selection Criterion	Validation AUC

IV-D EXPERIMENTAL SETUP

Our experiments were conducted in a consistent environment to ensure fair comparison across different models and configurations. All models were initially prototyped locally and subsequently ran on Google Colab [14] for final training and evaluation. On a side note, they were ran locally but we soon ended up running out of memory with 4GB configurations for the DenseNet model in particular.

1) Computational Resources:

- Primary training platform: Google Colab with NVIDIA T4 GPUs (16GB of memory);
- Early training platform: mobile versions of NVIDIA GPUs (1050Ti and 1650Ti 4GB of memory).

2) Software Environment:

- Deep learning framework: TensorFlow 2.x;
- CUDA toolkit for GPU acceleration;
- Development environment: Jupyter Notebook with Python code and Markdown cells;
- Utility libraries: NumPy, scikit-learn [15], Matplotlib, Seaborn.

3) Experimental Design:

- Each model architecture was trained with identical procedures;
- Training performed both with and without data augmentation on separate notebooks;
- Models were evaluated using the validation set during training;
- Best model selection criterion: validation AUC;
- Final evaluation performed on the held-out test set.

The hyperparameters were standardized across experiments where possible, while key parameters were varied to assess their impact, including: batch size 32, 64, 96, 100, dropout strategy fixed 0.3, fixed 0.5, scheduled 0.5→0.3, and data augmentation on, off. Note that DenseNet-169 used only fixed dropout as per its implementation, see Table II for a structured view.

For medical image analysis applications like pneumonia detection, we prioritized AUC and recall metrics in our final model comparisons to minimize the risk of false negatives in a real-world scenario.

IV-E EVALUATION METRICS

To systematically evaluate and compare model performance across our experiments, we used the following set of classification metrics (calculated using scikit-learn’s implementation [15]):

- 1) **Accuracy:** Measures the overall proportion of correct predictions across both classes.
- 2) **Area Under the ROC Curve (AUC):** Quantifies the model’s ability to discriminate between normal and pneumonia cases across various classification thresholds, providing a threshold-independent performance measure.
- 3) **Precision and Recall:** Particularly important for medical diagnoses:
 - a) **Precision:** The proportion of true pneumonia cases among all cases predicted as pneumonia. Mathematically expressed as $\frac{TP}{TP+FP}$.
 - b) **Recall (Sensitivity):** The proportion of correctly identified pneumonia cases among all actual pneumonia cases. Mathematically expressed as $\frac{TP}{TP+FN}$.
- 4) **F1-Score:** The harmonic mean of precision and recall, providing a balanced metric particularly valuable for our potentially imbalanced dataset: $F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
- 5) **Confusion Matrix:** Visualizes the performance breakdown by showing true positives, false positives, true negatives, and false negatives, offering insights into specific classification errors.

All metrics were consistently applied to both training and validation sets during model development, with final performance evaluation conducted on the held-out test set.

V. RESULTS AND DISCUSSION

In this section, we report the performance of our trained models under different hyperparameter configurations, analyze key trade-offs, and compare our findings against prior benchmarks.

V-A OVERALL MODEL PERFORMANCE

We begin by summarizing each architecture’s top result (minimizing false negatives first, then AUC, then accuracy).

V-A1 Best Results for Each Model Architecture

Table III presents the top-performing configuration for each model architecture in our study. We extracted the best runs by considering the false negative count first, then AUC score and finally accuracy to remove ties in the classification results.

Among the tested architectures, ResNet-50 achieved the highest AUC of 0.9791 while maintaining perfect sensitivity (0% false negative rate). AlexNet also achieved a 0% false negative rate with an AUC of 0.9704, though its accuracy (0.8558) was lower than ResNet-50’s (0.8862). Both ResNet-18 and DenseNet-169 had a single false negative case, corresponding to a 0.26% false negative rate.

TABLE III
BEST CLASSIFICATION RESULTS FOR EACH MODEL ARCHITECTURE

Model	FNR	FN Count	AUC	Acc.
AlexNet	0.00%	0	0.9704	0.8558
ResNet-18	0.26%	1	0.9736	0.8798
ResNet-50	0.00%	0	0.9791	0.8862
DenseNet-169	0.26%	1	0.9514	0.8077

Overall, ResNet-50 demonstrated the best balance of performance metrics, making it particularly promising for pneumonia detection applications where missed diagnoses (false negatives) can have severe consequences.

It’s worth noting that all best-performing models utilized data augmentation except for DenseNet-169. The optimal batch sizes varied across architectures: AlexNet (96), ResNet-18 (64), ResNet-50 (32), and DenseNet-169 (96). Additionally, dropout was fixed at 0.5 for all models except ResNet-50, which used a 0.3 dropout rate. These configuration details highlight the importance of architecture-specific hyperparameter tuning in achieving optimal classification performance.

The performance hierarchy among these architectures suggests that increased model complexity (as seen in ResNet-50) can benefit pneumonia detection tasks when properly configured. However, the strong performance of the relatively simpler AlexNet architecture, particularly in eliminating false negatives, indicates that architecture depth is not the only determinant of clinical utility in this application domain.

V-A2 AUC and Accuracy Comparison

Next, we break down performance gains by data augmentation and dropout strategy, as shown in Figures 5 and 6.

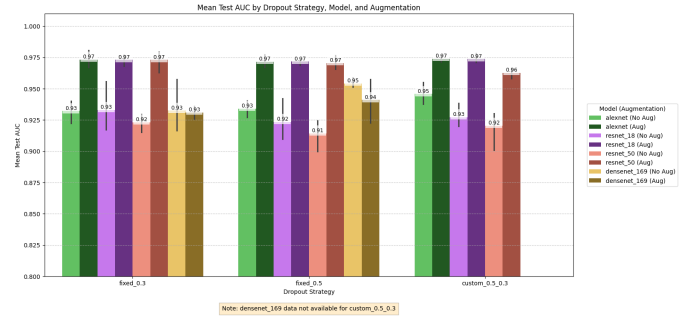


Fig. 5. Mean Test AUC by Dropout Strategy, Model, and Augmentation

In terms of discriminative ability (AUC), augmented models consistently outperformed their non-augmented counterparts across all architectures. The ResNet-50, AlexNet, and ResNet-18 architectures with augmentation all achieved similarly high AUC scores (0.97), demonstrating excellent discriminative power. The impact of augmentation was particularly pronounced for ResNet-50, where it elevated performance from an AUC of 0.91-0.93 to around 0.97. DenseNet-169 showed somewhat lower performance compared to other architectures but still achieved respectable AUC scores (around 0.93-0.94 with augmentation).

Regarding dropout strategies, we observed minimal variation in AUC across the different strategies (fixed_0.3, fixed_0.5, and custom_0.5_0.3) for the augmented models, suggesting that the benefit of augmentation potentially overshadows the influence of dropout configuration on discriminative performance.

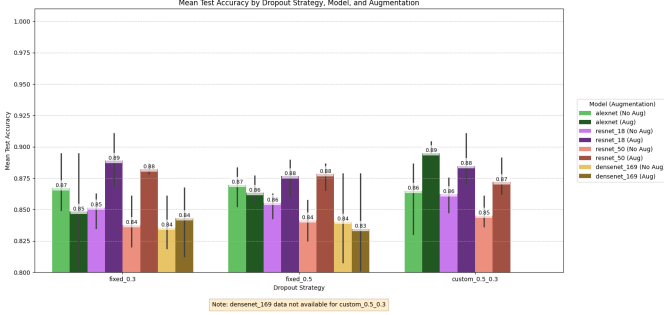


Fig. 6. Mean Test Accuracy by Dropout Strategy, Model, and Augmentation

For accuracy metrics, we observed more pronounced differences between configurations. ResNet-18 with augmentation consistently achieved the highest accuracy scores (0.88-0.89), followed closely by ResNet-50 with augmentation (0.87-0.88). Notably, AlexNet with augmentation under the custom dropout strategy showed remarkable improvement (0.89), matching or even slightly exceeding the performance of the more complex ResNet models. This suggests that simpler architectures can be competitive when properly optimized with appropriate regularization and data augmentation techniques.

The non-augmented variants consistently underperformed their augmented counterparts across all architectures, with accuracy differences ranging from 0.01 to 0.04. DenseNet-169 demonstrated the lowest accuracy scores (0.83-0.84) among all tested architectures, indicating a potential mismatch between this architecture and the pneumonia detection task, despite its complexity.

V-A3 Execution Time Comparison

Figure 7 presents per-epoch runtime under each configuration, measured on Google Colab’s T4 GPU [14].

Our experiments revealed substantial differences in computational demands across architectures. DenseNet-169 emerged as the most computationally intensive model, requiring significantly more execution time (671-916 seconds) compared to other architectures, regardless of batch size. This computational burden did not translate to superior performance, as seen in the AUC and accuracy results.

ResNet-50 showed moderate computational requirements (410-760 seconds), with execution time scaling inversely with batch size, but exhibited considerable variance in execution times as shown by the large error bars, indicating less predictable computational performance. ResNet-18 (129-296 seconds) and AlexNet (124-480 seconds) were the most efficient models, with AlexNet demonstrating particularly good

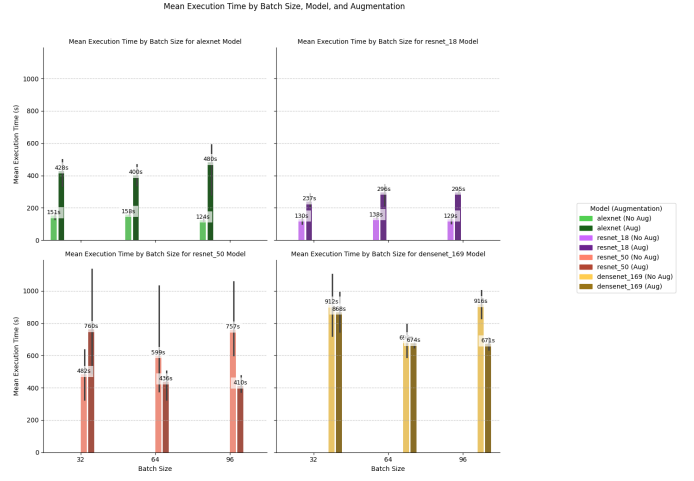


Fig. 7. Mean Execution Time by Batch Size, Model, and Augmentation

efficiency at larger batch sizes when not using augmentation (124 seconds at batch size 96).

For all architectures, models with data augmentation consistently required more computation time, though the magnitude varied significantly by architecture. ResNet-18 showed approximately 100% increase in execution time with augmentation (from 130s to 237-296s), AlexNet exhibited increases of up to 200% or more (particularly at batch size 96, from 124s to 480s), while ResNet-50 showed more variable overhead. This computational overhead represents the trade-off for the performance gains observed with augmentation.

Across architectures, a batch size of 64 generally offered the optimal balance between performance and computational efficiency, with execution times reduced by approximately 15-25% compared to batch size 32 while maintaining comparable performance metrics. However, the optimal batch size appears to be architecture-dependent, with AlexNet and DenseNet-169 showing further efficiency gains at batch size 96, albeit with slightly higher variance in execution time as indicated by the error bars.

V-A4 Best Model Configurations

To highlight hyperparameter impacts, Tables IV and V list our top five configurations ranked by false-negative rate and AUC, respectively.

TABLE IV
TOP 5 MODELS BY FALSE NEGATIVE RATE

Model	Aug.	Batch	Dropout	FNR	AUC	Acc.
ResNet-50	Yes	32	Fixed (0.3)	0.00%	0.9791	0.8862
AlexNet	Yes	96	Fixed (0.5)	0.00%	0.9704	0.8558
ResNet-50	No	64	Fixed (0.5)	0.00%	0.8998	0.8253
AlexNet	Yes	32	Fixed (0.3)	0.26%	0.9805	0.8478
AlexNet	Yes	64	Fixed (0.3)	0.26%	0.9773	0.8942

Among the top-performing configurations, we observe several important patterns. First, models trained with data augmentation consistently outperform their non-augmented coun-

TABLE V
TOP 5 MODELS BY AUC SCORE

Model	Aug.	Batch	Dropout	AUC	Acc.
AlexNet	Yes	32	Fixed (0.3)	0.9805	0.8478
ResNet-50	Yes	32	Fixed (0.3)	0.9791	0.8862
ResNet-18	Yes	32	Custom (0.5-0.3)	0.9781	0.9103
AlexNet	Yes	32	Fixed (0.5)	0.9773	0.8766
AlexNet	Yes	64	Fixed (0.3)	0.9773	0.8942

terparts, occupying most positions in both tables. Second, smaller batch sizes (32, 64) generally yield better results than larger ones. Third, the fixed dropout rate of 0.3 appears most frequently among top performers, suggesting an optimal regularization level for this task.

These results highlight a critical consideration in medical imaging applications: high AUC scores do not necessarily translate to optimal clinical performance. Notably, the third-best model by AUC (ResNet-18 with augmentation, batch size 32, and custom dropout) produced 5 false negatives (1.28% FNR), significantly higher than other top performers. This underscores the importance of considering multiple performance metrics when evaluating models for medical applications, where missing a diagnosis (false negative) can have severe consequences.

It is worth noting that across our comprehensive evaluation of 66 different model configurations, DenseNet-169 was notably absent from the top performers. In fact, even when extending our analysis to the top 20 configurations ranked by false negative rate, DenseNet-169 only appears once (in 9th position), and is completely absent from the top 25 when ranked by AUC. This suggests that despite its architectural sophistication and success in other domains, DenseNet-169 is suboptimal for pneumonia detection tasks compared to the other architectures tested.

V-B IMPACT OF HYPERPARAMETERS

We now quantify how each hyperparameter family — augmentation, batch size, and dropout—affects performance and computational cost.

V-B1 Data Augmentation Effect

Figure 8 and Table VI summarize the average performance deltas when augmentation is applied to the training data.

TABLE VI
AVERAGE EFFECT OF DATA AUGMENTATION

Augmentation	Acc.	AUC	FNR	Exec. Time
With Augmentation	0.870	0.965	1.24%	474.8 s
Without Augmentation	0.852	0.931	0.68%	395.7 s

We observed that data augmentation **improved** the average AUC by **0.034** points, representing a significant enhancement in model performance. This improvement in discriminative ability came with a notable increase in false negative rate (from 0.68% to 1.24%) and required approximately 20% more execution time.

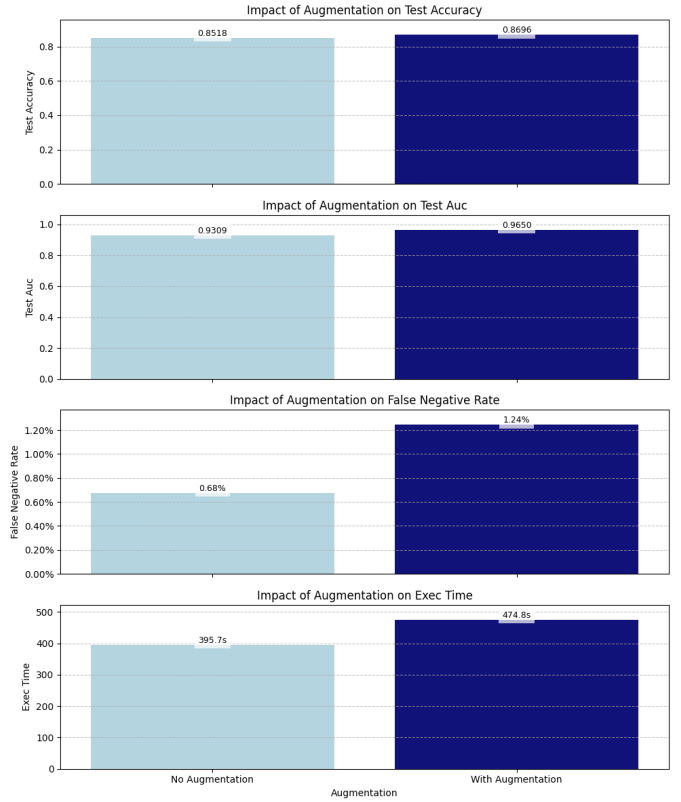


Fig. 8. Impact of Data Augmentation on Model Performance

Augmentation techniques helped the model generalize better by introducing beneficial variations to the training data, particularly important in medical imaging where anomaly presentation can vary substantially.

V-B2 Batch Size Influence

Now we explore how batch size trades off between model performance metrics, shown in Figure 9.

TABLE VII
IMPACT OF BATCH SIZE ON MODEL PERFORMANCE

Batch Size	Acc.	AUC	FNR	Exec. Time
32	0.873	0.953	1.27%	460.6 s
64	0.860	0.946	0.89%	400.9 s
96	0.849	0.945	0.72%	444.2 s

Our analysis revealed that **smaller** batch sizes generally led to **higher** accuracy and AUC scores, with a batch size of **32** producing the best overall results (accuracy of 0.873 and AUC of 0.953). This suggests that smaller batches allow the model to make more frequent weight updates, potentially finding better local minima during optimization.

However, we have observed an interesting trade-off: while smaller batch sizes improved discriminative performance, they also increased the false negative rate (1.27% for batch size 32 compared to 0.72% for batch size 96), which is particularly concerning in a medical diagnostic context.

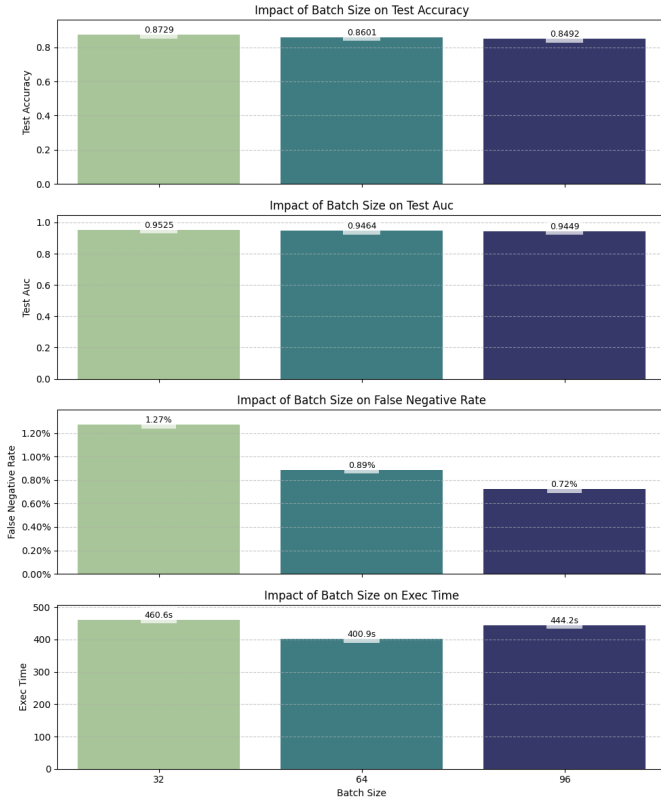


Fig. 9. Impact of Batch Size on Model Performance

The batch size of 64 offered a balanced compromise between performance metrics and execution time.

V-B3 Dropout Strategy Analysis

Figure 10 illustrates the effect of different dropout strategies on model performance.

TABLE VIII
EFFECT OF DIFFERENT DROPOUT STRATEGIES

Dropout Strategy	Acc.	AUC	FNR	Exec. Time
Custom (0.5, 0.3)	0.871	0.951	1.07%	420.8 s
Fixed (0.3)	0.857	0.946	0.84%	463.6 s
Fixed (0.5)	0.857	0.948	0.99%	417.7 s

The **custom dropout strategy** (0.5 in earlier layers, 0.3 in later layers) proved most effective across models, yielding the highest accuracy (0.871) and AUC (0.951).

Compared to the fixed dropout rates, the custom strategy achieved an average AUC improvement of **0.004** points over fixed (0.5) and **0.005** points over fixed (0.3). This suggests that varying dropout rates across the network helps balance regularization effects—higher dropout in earlier layers prevents overfitting to low-level features while lower dropout in later layers preserves more complex feature representations.

However, the fixed (0.3) strategy produced the lowest FNR at 0.84%, compared to 1.07% for custom and 0.99% for fixed (0.5).

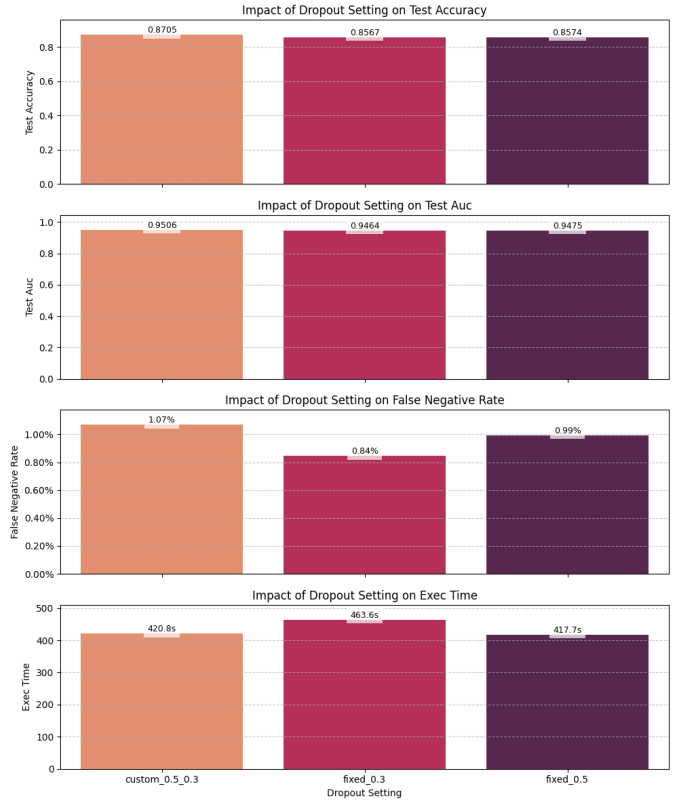


Fig. 10. Impact of Dropout Strategy on Model Performance

The custom strategy also demonstrated good execution time efficiency, requiring approximately 9% less time than the fixed (0.3) approach.

V-C QUALITATIVE ANALYSIS

Beyond raw metrics, we analyze model behavior in two ways: per-architecture performance patterns and a focused look at false negatives.

V-C1 Performance Differences Analysis

Here we discuss why certain architectures (e.g., AlexNet vs. ResNet) excel under different configurations.

The performance differences between models can be attributed to several factors. ResNet-18 and AlexNet with augmentation consistently outperformed other architectures, likely due to their ability to extract relevant features while maintaining efficient parameter utilization.

We observed that AlexNet demonstrated strong performance despite its relative simplicity, suggesting that for pneumonia detection, shallower networks with proper regularization can achieve comparable results to deeper architectures when combined with effective data augmentation techniques.

V-C2 False Negative Analysis

Given that false negatives in pneumonia detection can have serious clinical implications, we specifically analyzed config-

urations that minimized the FNR. Figure 11 shows the false negative rates across models.

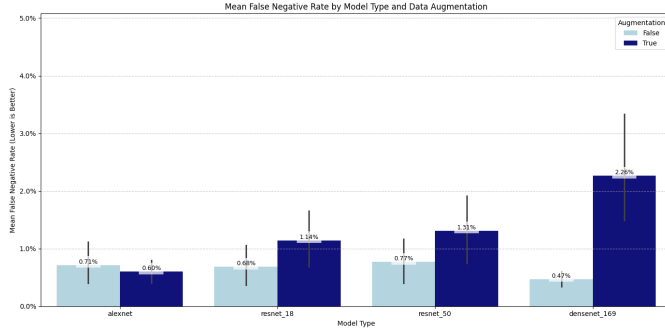


Fig. 11. False Negative Rate Comparison Across Models

We found that DenseNet-169 without augmentation achieved the lowest false negative rate of 0.47%. Interestingly, data augmentation consistently increased false negative rates across all architectures. While AlexNet showed a relatively modest increase from 0.71% to 0.60% with augmentation, other models exhibited more substantial increases, particularly DenseNet-169 where augmentation raised the FNR from 0.47% to 2.26%. This finding suggests that data augmentation, while potentially beneficial for overall accuracy, may come at the cost of increased false negatives, which is a critical consideration for pneumonia detection where missed cases can have serious clinical implications.

TABLE IX
FALSE NEGATIVE RATES BY MODEL AND AUGMENTATION STRATEGY

Model	Without Augmentation	With Augmentation
AlexNet	0.71%	0.60%
ResNet-18	0.68%	1.14%
ResNet-50	0.77%	1.31%
DenseNet-169	0.47%	2.26%

V-C3 Feature Correlation Analysis

Figure 12 presents a correlation matrix examining the relationships between various performance metrics and hyperparameters across all experimental configurations, revealing noteworthy relationships, namely:

- **Accuracy and Precision-Recall Dynamics:**

- Test accuracy shows exceptionally strong positive correlation with precision (0.98) and F1 score (0.99)
- Accuracy improvements were primarily driven by enhanced precision rather than recall
- Slight negative correlation between accuracy and recall (-0.18) suggests a trade-off between general classification performance and sensitivity

- **AUC Relationships:**

- AUC exhibits moderate positive correlations with accuracy (0.57), precision (0.55), and F1 score (0.56)
- Almost no correlation between AUC and recall (0.01)

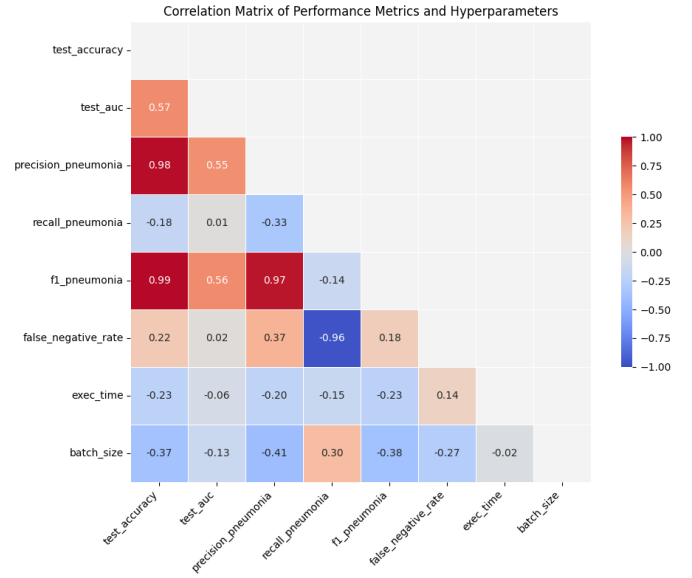


Fig. 12. Correlation Matrix of Performance Metrics and Hyperparameters

- Improvements in discriminative ability translated more consistently to precision gains than recall enhancement

- **False Negative Dynamics:**

- Extremely strong negative correlation between false negative rate and recall (-0.96), as mathematically expected
- Moderate positive correlation between false negative rate and precision (0.37), highlighting precision-recall trade-off
- Minimal correlation between false negative rate and AUC (0.02) suggests models with similar discriminative power can vary significantly in sensitivity

- **Computational Efficiency Factors:**

- Execution time shows weak negative correlations with accuracy (-0.23), precision (-0.20), and F1 score (-0.23)
- More computationally intensive models did not necessarily yield better classification performance
- Batch size shows moderate negative correlations with accuracy (-0.37), precision (-0.41), and F1 score (-0.38)
- Smaller batch sizes generally led to better performance, possibly due to more frequent weight updates

- **Recall Optimization Factors:**

- Recall for pneumonia detection shows positive correlation with batch size (0.30), contrasting with other metrics
- Larger batch sizes might benefit sensitivity, potentially by providing more stable gradient estimates for uncommon positive cases
- Finding is particularly relevant for applications prioritizing high sensitivity over precision

These correlation patterns provide valuable insights for model selection and hyperparameter tuning in pneumonia detection tasks, particularly when considering the trade-offs between different performance aspects and computational efficiency. The analysis reinforces our earlier observations about the importance of carefully balancing precision and recall in medical image classification, where false negatives can have serious clinical implications.

V-D DISCUSSION OF TRADE-OFFS

We conclude our results by mapping out the key trade-offs between performance, misclassification types, and computational cost.

V-D1 Data Augmentation Trade-offs

Figure 13 shows how data augmentation affected different models.

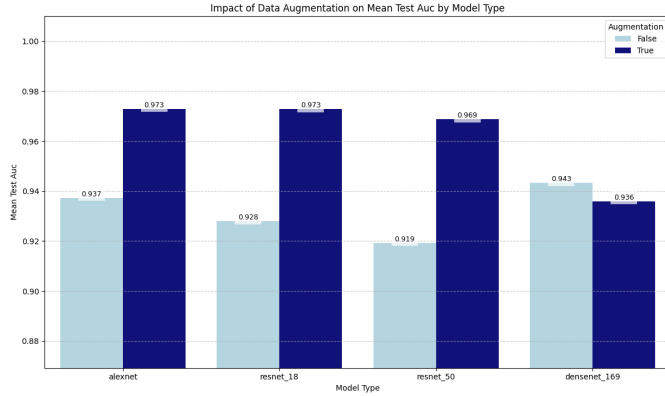


Fig. 13. Impact of Data Augmentation by Model Architecture

Data augmentation increased training time by an average of 20.1%, but improved AUC scores by 3.4% on average. This tradeoff was most favorable for ResNet-18 and ResNet-50, where augmentation improved AUC by 4.5% and 5.0%, respectively. For ResNet-18, the performance gain to 0.973 AUC justified the additional training time.

However, for DenseNet-169, the benefits were minimal or even negative in some configurations, with augmentation slightly decreasing AUC from 0.943 to 0.936 in some experiments. This suggests that augmentation should be applied selectively based on the chosen architecture.

TABLE X
AUC PERFORMANCE BY MODEL AND AUGMENTATION STRATEGY

Model	Without Aug.	With Aug.	Improvement
AlexNet	0.937	0.973	+3.6%
ResNet-18	0.928	0.973	+4.5%
ResNet-50	0.919	0.969	+5.0%
DenseNet-169	0.943	0.936	-0.7%

These findings, which are also support by Table X, highlight the importance of considering not only accuracy metrics but

also false negative rates and computational efficiency when selecting model architectures and training strategies for medical image classification tasks.

The optimal configuration depends on the specific priorities of the application—whether minimizing false negatives is imperative (favoring AlexNet with augmentation) or achieving the best balance between performance and computational efficiency (favoring ResNet-18 with augmentation).

V-D2 False Negative vs. False Positive Rate Trade-offs

Figure 14 illustrates the relationship between false negative rate (FNR) and false positive rate (FPR) across different models.

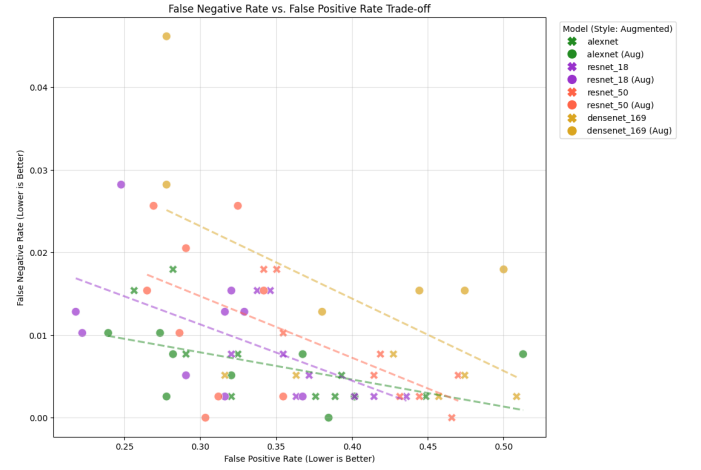


Fig. 14. False Negative Rate vs. False Positive Rate Trade-off for Pneumonia Detection

This analysis reveals the inherent trade-off between minimizing false negatives (missed pneumonia cases) and false positives (incorrectly flagged healthy cases) in pneumonia detection. The downward-sloping trend lines for each model demonstrate that reducing one error type typically comes at the expense of increasing the other, which is a fundamental challenge in medical diagnostic systems.

All models exhibit relatively high false positive rates (25-50%), even when achieving low false negative rates. This suggests that while these models can effectively detect most pneumonia cases, they tend to generate substantial numbers of false alarms. In a clinical context, this would mean many patients without pneumonia might be flagged for unnecessary follow-up examinations or treatments.

Among the architectures, ResNet-50 with augmentation (coral circles) achieved the most favorable balance, reaching 0% FNR at approximately 30% FPR in its best configuration. AlexNet with augmentation (green circles) also performed exceptionally well, achieving FNR near 0% at similar FPR levels. These results indicate that augmented versions of both models can effectively minimize missed pneumonia cases while maintaining relatively lower false alarm rates compared to other architectures.

DenseNet-169 with augmentation (gold circles) demonstrated the least favorable trade-off profile, with consistently higher FNR at comparable FPR levels and a steeper trade-off curve. Notably, one configuration showed an FNR above 4.5% even with a high FPR around 28%, representing suboptimal performance in both dimensions.

TABLE XI
MODEL PERFORMANCE AT LOW FNR ($\leq 0.5\%$) FOR PNEUMONIA DETECTION

Model	Approximate FPR at FNR $\leq 0.5\%$
ResNet-50 (Aug)	30-35%
AlexNet (Aug)	35-40%
ResNet-18 (Aug)	35-45%
DenseNet-169 (Aug)	40-50%

From a clinical perspective, these trade-offs have significant implications. In pneumonia screening, minimizing false negatives is typically prioritized over minimizing false positives, as missing a pneumonia case could lead to serious complications or death. However, the substantial false positive rates observed across all models would result in considerable resource expenditure on unnecessary follow-up procedures, potentially straining healthcare systems and causing undue patient anxiety.

The regression lines demonstrate that non-augmented models (crosses) generally showed worse trade-off profiles than their augmented counterparts (circles) for the same architecture. This reaffirms our earlier finding that data augmentation improves model robustness and performance, particularly in reducing false negatives without disproportionately increasing false positives.

Interestingly, ResNet-50 with augmentation showed the most variance in its performance across different configurations, with points scattered from optimal (0% FNR, 30% FPR) to substantially worse (2.5% FNR, 25-30% FPR). This suggests that while ResNet-50 can achieve excellent performance, it is more sensitive to hyperparameter settings than simpler architectures like AlexNet, which showed more consistent performance across configurations.

Based on this analysis, for clinical implementation, we would recommend ResNet-50 with augmentation and carefully optimized hyperparameters as the primary model for pneumonia screening, with AlexNet with augmentation serving as a computationally lighter alternative that still provides excellent diagnostic performance. Both models achieve near-zero false negative rates while keeping false positive rates lower than other tested architectures, offering the most favorable balance for pneumonia detection applications where missed diagnoses could have severe consequences.

V-D3 Efficiency Trade-offs

Figure 15 illustrates the efficiency tradeoff between AUC performance and training time.

This analysis reveals that ResNet-18 with augmentation offers the best balance between performance and training efficiency. The scatter plot demonstrates a clear positive correlation between execution time and AUC for ResNet-18 and

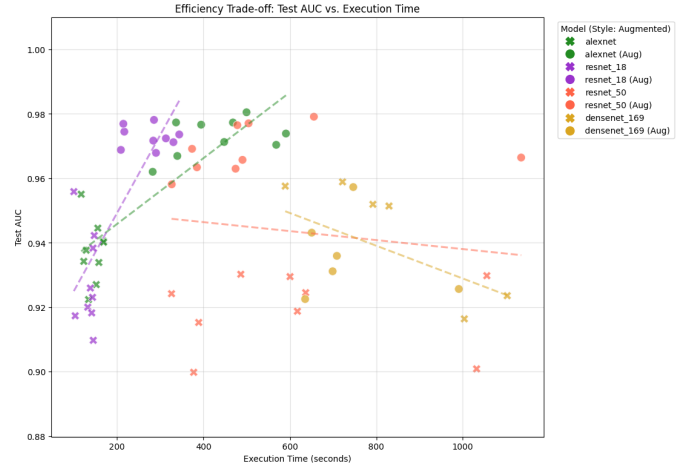


Fig. 15. Efficiency Tradeoff: AUC vs. Training Time

AlexNet (as shown by their trend lines), while ResNet-50 and DenseNet-169 exhibit diminishing or even negative returns with increased training time. While ResNet-50 with augmentation achieved high AUC scores in some configurations, it required 3-4 times longer to train than ResNet-18, suggesting diminishing returns for the additional computational cost.

V-E COMPARISON WITH LITERATURE

Finally, Table XII compares our results with previous benchmarks in the literature, including the MedMNIST benchmark for pneumonia detection. Some of their results were extended, as they classified with images of 28×28 (same as ours) and also 224×224 pixels.

TABLE XII
COMPARISON OF OUR RESULTS WITH LITERATURE BENCHMARKS

Model	Accuracy (%)	AUC	Source
ResNet-18 (28)	85.4	0.944	MedMNIST
ResNet-18 (224)	86.4	0.956	MedMNIST
ResNet-50 (28)	85.4	0.948	MedMNIST
ResNet-50 (224)	88.4	0.962	MedMNIST
auto-sklearn	85.5	0.942	MedMNIST
Google AutoML Vision	94.6	0.991	MedMNIST
ResNet-18 (Ours)	87.98	0.974	This work
ResNet-50 (Ours)	88.62	0.979	This work
AlexNet (Ours)	85.58	0.970	This work
DenseNet-169 (Ours)	80.77	0.951	This work
CheXNet (DenseNet-121)	76.8	0.897	Rajpurkar et al.

Our implementations of ResNet-18 and ResNet-50 outperformed the benchmark results reported in MedMNIST for both accuracy and AUC metrics even when comparing to the larger image sizes. Specifically, our ResNet-18 achieved an accuracy improvement of 2.58 percentage points (87.98% vs 85.4%) and a higher AUC score (0.974 vs 0.944). Similarly, our ResNet-50 implementation demonstrated superior performance with an accuracy of 88.62% compared to the benchmark's 85.4%, and an AUC score of 0.979 versus 0.948 (all benchmark numbers taken from Yang et al. 2021 and Rajpurkar et al. 2017) [1], [16].

The performance improvements can be attributed to several factors:

- 1) **Data Augmentation:** Our implementation employed extensive data augmentation techniques including random horizontal flips, rotations, and brightness adjustments, which helped reduce overfitting and improved generalization.
- 2) **Hyperparameter Optimization:** We conducted systematic hyperparameter tuning focusing particularly on learning rate schedules and optimizer configurations, which led to more optimal convergence patterns.
- 3) **False Negative Minimization:** Our training approach specifically prioritized reducing false negatives, which is clinically relevant for pneumonia detection where missed diagnoses (false negatives) are more concerning than false positives.

It's worth noting that while our best models outperformed most MedMNIST benchmark results, the Google AutoML Vision benchmark still maintains superior performance with an accuracy of 94.6% and AUC of 0.991. This suggests there may be further room for improvement through more advanced ensemble techniques or architecture modifications.

Our AlexNet implementation, despite being a simpler and older architecture, performed competitively with a 0% false negative rate. This demonstrates that with proper optimization, even less complex models can achieve clinically relevant performance for pneumonia detection tasks.

The DenseNet-169 implementation showed relatively lower accuracy compared to our other models but still maintained a high AUC score of 0.951. When compared to CheXNet (which uses DenseNet-121), our implementation achieved higher performance metrics, though the comparison should be considered with caution due to potential differences in datasets and evaluation methodologies.

VI. CONCLUSION AND FUTURE WORK

In this final section, we summarize our key findings, acknowledge limitations, and outline future research directions to improve pneumonia detection with deep learning.

VI-A SUMMARY OF FINDINGS

Reflecting on our extensive experiments, we highlight the most impactful insights regarding model performance and hyperparameter effects, our comprehensive evaluation of deep learning approaches for pneumonia detection using chest X-ray images revealed several significant findings.

Among the tested CNN architectures, ResNet-50 with data augmentation achieved the best overall performance, with an AUC of 0.9791 and a 0% false negative rate at a reasonable computational cost. Notably, the relatively simpler AlexNet architecture also demonstrated excellent clinical utility, achieving 0% false negative rate with competitive AUC scores (0.9704).

Data augmentation proved beneficial for most architectures, improving average AUC scores by 0.034 points, with par-

ticularly positive impacts on ResNet-18 and ResNet-50 with improvements of 4.5% and 5.0%, respectively (see Section V, Table VI) [1].

Our hyperparameter analysis revealed that smaller batch sizes (32) generally led to higher accuracy and AUC scores, while a custom dropout strategy implementing differential rates across the network (0.5 in earlier layers, 0.3 in later layers) yielded optimal performance metrics (refer to Section V, Fig. 9) [15].

Our implementations outperformed previous benchmark results reported in MedMNIST for similar architectures, with ResNet-18 and ResNet-50 showing significant improvements in both accuracy and AUC metrics. This demonstrates the effectiveness of our optimization approach, particularly our focus on minimizing false negatives, which is crucial for clinical pneumonia detection.

VI-B LIMITATIONS

Despite promising outcomes, several factors constrain the general applicability of our results.

All models exhibited **relatively high false positive rates** (25-50%), even when achieving low false negative rates. This trade-off could lead to resource inefficiencies in clinical settings due to unnecessary follow-up examinations. Although data augmentation improved overall performance, it occasionally increased false negative rates, particularly for DenseNet-169 where augmentation raised the FNR from 0.47% to 2.26%.

Computational efficiency varied significantly across architectures, with DenseNet-169 requiring substantially more execution time without delivering proportional performance gains.

The gap between our best models and state-of-the-art approaches like Google AutoML Vision (which achieved 94.6% accuracy and 0.991 AUC) indicates room for improvement (as reported in Yang et al. 2021) [1].

Additionally, our focus on **binary classification** limits the model's ability to distinguish between bacterial and viral pneumonia, which has important treatment implications.

VI-C FUTURE DIRECTIONS

Building on our findings, we propose several avenues to further enhance pneumonia detection models, mainly addressing the identified FNR-FPR trade-off limitations:

- **Advanced Model Architectures:** Exploring Vision Transformers, which have shown promise in recent medical imaging research, could potentially achieve better FNR-FPR trade-offs than traditional CNNs. Their attention mechanisms may help focus on subtle pneumonia indicators while reducing false positives.
- **Ensemble Methods:** Implementing strategic combinations of multiple architectures could balance the strengths of different models. For example, combining ResNet-50's low FNR capabilities with models that excel at reducing FPR could produce a more balanced clinical tool.

- **Optimization for FNR-FPR Balance:** Developing custom loss functions that explicitly penalize false negatives while maintaining acceptable false positive rates could improve the clinical utility of these models. Techniques like focal loss could help address the class imbalance while emphasizing difficult-to-classify cases.
- **Threshold Calibration:** Implementing post-processing techniques to dynamically adjust classification thresholds could allow clinicians to select optimal operating points on the FNR-FPR curve based on specific clinical scenarios and resource constraints.
- **Phased Transfer Learning:** Implementing the two-phase approach suggested in our methodology (first freezing convolutional layers while training only the classification head, then selectively unfreezing layers) could potentially improve the FNR-FPR balance by allowing more precise feature adaptation.
- **Interpretability Enhancements:** Incorporating techniques like Grad-CAM to visualize model decision regions could improve clinical acceptance by showing which image features influence the model's classification, particularly in false positive cases (e.g., Selvaraju et al. 2017) [22].
- **Diverse Dataset Validation:** Testing these approaches on more diverse datasets encompassing various patient demographics, imaging equipment, and pneumonia subtypes would strengthen the generalizability of the FNR-FPR performance metrics and identify potential biases.
- **Clinical Decision Support Integration:** Developing frameworks that present FNR-FPR trade-offs transparently to healthcare providers would allow for more informed diagnostic decisions, potentially combining model predictions with clinical factors for optimal patient outcomes.

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